Population Attributable Fraction of Obesity for Hypercapnic Respiratory Failure

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```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   3.5.2     v tibble    3.2.1
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.0.4
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(haven)
library(janitor)
```

```
Attaching package: 'janitor'

The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```r
library(gtsummary)
library(ggplot2)


rm(list = ls())
#stata_data <- readRDS("C:/Users/ajsch/OneDrive/Desktop/full_db.rds")
stata_data <- readRDS('/Users/blocke/Box Sync/Residency Personal Files/Scholarly Work/Locke Research Projects/TriNetX Code/data_

nrow(stata_data)
```

```
[1] 833476
```

```r
library(haven)
library(dplyr)

# Convert labelled variables to numeric properly
stata_data <- stata_data %>%
  mutate(
    niv_proc = as.numeric(as_factor(niv_proc)) - 1,  # converts 1="Yes", 2="No" to 0/1
    imv_proc = as.numeric(as_factor(imv_proc)) - 1,
    highest_paco2 = as.numeric(highest_paco2),
    abg_ph = as.numeric(abg_ph),
    vbg_co2 = as.numeric(vbg_co2),
    vbg_ph = as.numeric(vbg_ph)
  )
```

Below is my first (probably most stringent) iteration of hypercapnic respiratory failure: co2 greater than 45 if abg, 50 if vbg, ph <7.35 - they have to have either an abg or a vbg showing this, AND either NIV or IMV or both. I've written it so that ABG values and VBG values cannot mix

```
stata_data <- stata_data %>%
  mutate(
    gas_source = case_when(
      !is.na(highest_paco2) & !is.na(abg_ph) ~ "ABG",
      TRUE ~ "VBG"
    )
  )
```

```
stata_data <- stata_data %>%
  mutate(
    hrf_hypercapnic = case_when(
      # ABG-based HRF
      (niv_proc == 1 | imv_proc == 1) &
        gas_source == "ABG" &
        highest_paco2 >= 45 &
        abg_ph <= 7.35 ~ 1,

      # VBG-based HRF (only when ABG incomplete)
      (niv_proc == 1 | imv_proc == 1) &
        gas_source == "VBG" &
        !is.na(vbg_co2) & !is.na(vbg_ph) &
        vbg_co2 >= 50 &
        vbg_ph <= 7.35 ~ 1,

      TRUE ~ 0
    ),
    hrf_hypercapnic = factor(hrf_hypercapnic, labels = c("No", "Yes"))
  )
```

Counts (bottom right is amount of patients meeting criteria above, adds up to full dataset)

```r
table(
  hrf = stata_data$hrf_hypercapnic,
  any_vent = (stata_data$niv_proc == 1 | stata_data$imv_proc == 1)
)
```

```
     any_vent
hrf     FALSE   TRUE
  No   747798  59904
  Yes       0  25774
```

Counts for categories of obesity:

```r
stata_data <- stata_data %>%
  mutate(
    curr_bmi_num = suppressWarnings(
      as.numeric(na_if(as.character(zap_labels(curr_bmi)), "n/a"))
    )
  )
```

```r
stata_data <- stata_data %>%
mutate(
bmi_category = case_when(
is.na(curr_bmi_num) ~ NA_character_,
curr_bmi_num < 18.5 ~ "Underweight",
curr_bmi_num < 25   ~ "Normal weight",
curr_bmi_num < 30   ~ "Overweight",
curr_bmi_num < 35   ~ "Obesity I",
curr_bmi_num < 40   ~ "Obesity II",
curr_bmi_num >= 40  ~ "Obesity III"
)
)
```

```r
table(stata_data$hrf_hypercapnic, useNA = "ifany")
```

```
    No    Yes
807702  25774
```

```r
summary(stata_data$curr_bmi_num)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
  10.09   26.31   33.34   33.64   41.20   50.00  450715
```

```r
table(stata_data$bmi_category, useNA = "ifany")
```

```
Normal weight     Obesity I    Obesity II   Obesity III    Overweight
        65912         58317         44022        130477         74499
  Underweight          <NA>
         9534        450715
```

Counts for both HRF as defined above AND current BMI above 30

```r
stata_data %>%
  filter(
    hrf_hypercapnic == "Yes",
    curr_bmi_num > 30
  ) %>%
  summarise(
    n_patients = n()
  )
```

n_patients
4092

Counts for HRF and BMI by category

```r
stata_data %>%
  filter(
    hrf_hypercapnic == "Yes",
    !is.na(curr_bmi_num)
  ) %>%
  mutate(
    bmi_category = case_when(
      curr_bmi_num < 18.5 ~ "Underweight",
      curr_bmi_num < 25   ~ "Normal",
      curr_bmi_num < 30   ~ "Overweight",
      curr_bmi_num < 35   ~ "Obesity I",
      curr_bmi_num < 40   ~ "Obesity II",
      curr_bmi_num >= 40  ~ "Obesity III"
    )
  ) %>%
  count(bmi_category) %>%
  mutate(pct = round(100 * n / sum(n), 1))
```

bmi_category	n	pct
Normal	2311	25.1
Obesity I	1766	19.2
Obesity II	1131	12.3
Obesity III	1202	13.1
Overweight	2287	24.9
Underweight	503	5.5

Below is regression, starting with BMI (the degrees of freedom is 112k which seems appropriate - includes all patients with a BMI and labs/vent indicating NO HRF, plus the patients with BMI and hypercapnic respiratory failure as defined above)

```r
stata_hrf_analysis <- stata_data %>%
  filter(!is.na(curr_bmi_num),
         (!is.na(highest_paco2) & !is.na(abg_ph)) |
         (!is.na(vbg_co2) & !is.na(vbg_ph)))
```

```r
logit_bmi_hrf <- glm(hrf_hypercapnic ~ curr_bmi_num,
data = stata_hrf_analysis,
family = binomial)

summary(logit_bmi_hrf)
```

```
Call:
glm(formula = hrf_hypercapnic ~ curr_bmi_num, family = binomial,
    data = stata_hrf_analysis)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  -2.92776    0.04482  -65.33   <2e-16 ***
curr_bmi_num  0.01750    0.00146   11.98   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 63611  on 112008  degrees of freedom
Residual deviance: 63470  on 112007  degrees of freedom
AIC: 63474

Number of Fisher Scoring iterations: 5
```

```r
exp(cbind(OR = coef(logit_bmi_hrf), confint(logit_bmi_hrf)))
```

```
Waiting for profiling to be done...

                    OR      2.5 %     97.5 %
(Intercept)  0.05351677 0.04901251 0.05842539
curr_bmi_num 1.01765221 1.01474067 1.02056488
```

OR shows small increase in risk with BMI, though is statistically significant

Regression done again adjusted for variables NOT influenced by BMI (age, sex, ethnicity included - didn't include location as it appears to be the same for all entries)

```
baseline_model <- glm(
hrf_hypercapnic ~ curr_bmi_num + age_at_encounter + sex + ethnicity,
data = stata_hrf_analysis,
family = binomial
)

summary(baseline_model)
```

```
Call:
glm(formula = hrf_hypercapnic ~ curr_bmi_num + age_at_encounter +
    sex + ethnicity, family = binomial, data = stata_hrf_analysis)

Coefficients:
                             Estimate Std. Error z value Pr(>|z|)
(Intercept)                -3.3143678  0.0653822 -50.692  < 2e-16 ***
curr_bmi_num                0.0194994  0.0014798  13.177  < 2e-16 ***
age_at_encounter            0.0028333  0.0006493   4.363 1.28e-05 ***
sexMale                     0.2388572  0.0222256  10.747  < 2e-16 ***
ethnicityHispanic or Latino -0.0444911  0.0432380  -1.029    0.303
ethnicityUnknown            0.1622532  0.0302736   5.360 8.34e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 63611  on 112008  degrees of freedom
Residual deviance: 63302  on 112003  degrees of freedom
AIC: 63314

Number of Fisher Scoring iterations: 5
```

```r
exp(cbind(OR = coef(baseline_model), confint(baseline_model)))
```

```
Waiting for profiling to be done...
```

	OR	2.5 %	97.5 %
(Intercept)	0.03635703	0.03197381	0.04131457
curr_bmi_num	1.01969074	1.01673425	1.02264945
age_at_encounter	1.00283727	1.00156429	1.00411687
sexMale	1.26979724	1.21572690	1.32639886
ethnicityHispanic or Latino	0.95648412	0.87803709	1.04023921
ethnicityUnknown	1.17615801	1.10805001	1.24767299

Association appears to be stronger once adjusted

Just out of curiosity ran Chi square analysis below

```r
bmi_hrf_subset <- stata_hrf_analysis %>%
filter(!is.na(bmi_category))
```

```r
bmi_hrf_summary <- bmi_hrf_subset %>%
group_by(bmi_category) %>%
summarise(
n_total = n(),
n_hrf = sum(hrf_hypercapnic == "Yes"),
n_no_hrf = sum(hrf_hypercapnic == "No"),
pct_hrf = round(100 * n_hrf / n_total, 1),
pct_no_hrf = round(100 * n_no_hrf / n_total, 1)
) %>%
arrange(factor(bmi_category, levels = c("Underweight", "Normal weight", "Overweight",
"Obesity I", "Obesity II", "Obesity III")))

chisq_test <- chisq.test(table(bmi_hrf_subset$bmi_category, bmi_hrf_subset$hrf_hypercapnic))

bmi_hrf_summary
```

bmi_category	n_total	n_hrf	n_no_hrf	pct_hrf	pct_no_hrf
Underweight	5061	503	4558	9.9	90.1
Normal weight	31060	2311	28749	7.4	92.6
Overweight	31756	2287	29469	7.2	92.8
Obesity I	21692	1766	19926	8.1	91.9
Obesity II	12223	1131	11092	9.3	90.7
Obesity III	10217	1202	9015	11.8	88.2

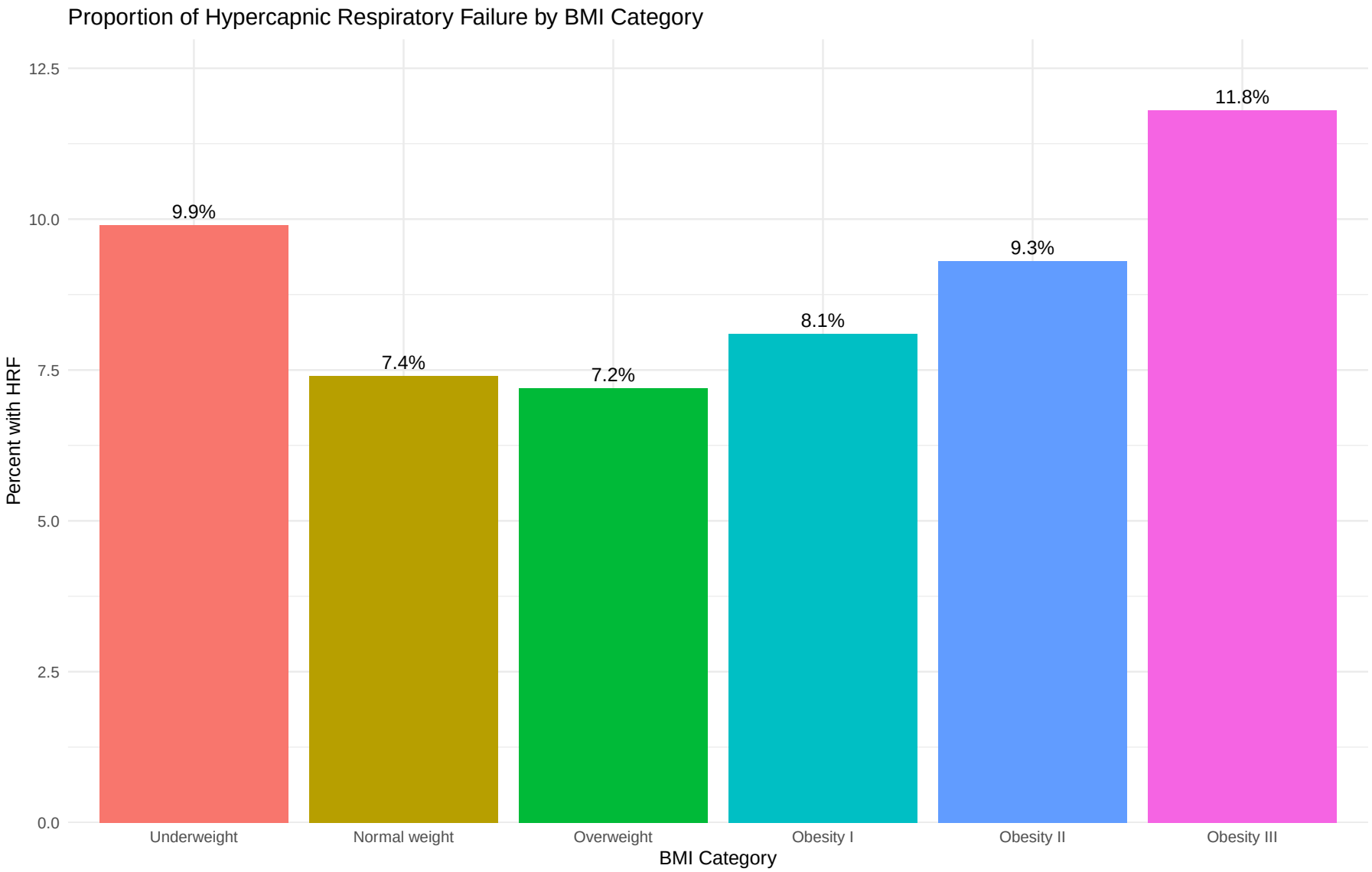
```
chisq_test$p.value
```

```
[1] 1.243203e-57
```

```
library(ggplot2)

# Make sure bmi_category is ordered
bmi_hrf_summary <- bmi_hrf_summary %>%
  mutate(bmi_category = factor(bmi_category,
                               levels = c("Underweight", "Normal weight", "Overweight",
                                          "Obesity I", "Obesity II", "Obesity III")))

# Plot proportion of HRF by BMI category
ggplot(bmi_hrf_summary, aes(x = bmi_category, y = pct_hrf, fill = bmi_category)) +
  geom_col() +
  geom_text(aes(label = paste0(pct_hrf, "%")), vjust = -0.5) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.1))) +
  labs(
    title = "Proportion of Hypercapnic Respiratory Failure by BMI Category",
    x = "BMI Category",
    y = "Percent with HRF"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```



Second analysis:

Regression done with new definition of hypercapnic respiratory failure - pH and CO2 definition as above, except now including (required) ICD codes instead of ventilation (I only used 4 - the first four that are listed in database):

```r
stata_data <- stata_data %>%
  mutate(
    # Lab-based hypercapnic respiratory failure (ABG or VBG)
    hrf_hypercapnic_labs_v2 = case_when(
      !is.na(highest_paco2) & !is.na(abg_ph) &
       highest_paco2 >= 50 & abg_ph < 7.35 ~ 1,

      !is.na(vbg_co2) & !is.na(vbg_ph) &
        vbg_co2 >= 50 & vbg_ph < 7.35 ~ 1,

      TRUE ~ 0
    ),

    # ICD-based hypercapnic respiratory failure
    hrf_hypercapnic_icd_v2 = ifelse(
      has_j9612 == 1 |
      has_j9622 == 1 |
      has_j9692 == 1 |
      has_j9602 == 1,
      1, 0
    ),

    # Final combined HRF definition (labs OR ICD)
    hrf_hypercapnic_v2 = ifelse(
      hrf_hypercapnic_labs_v2 == 1 |
      hrf_hypercapnic_icd_v2 == 1,
      1, 0
    )
  )
```

```r
stata_data <- stata_data %>%
  mutate(
    hrf_hypercapnic_v2_strict = ifelse(
      hrf_hypercapnic_labs_v2 == 1 &
      hrf_hypercapnic_icd_v2  == 1,
      1, 0
```

```
    )
  )
```

```
table(stata_data$hrf_hypercapnic_v2_strict, useNA = "ifany")
```

```
     0      1
818503  14973
```

```
stata_hrf_strict_analysis <- stata_data %>%
  mutate(
    has_gas = (!is.na(highest_paco2) & !is.na(abg_ph)) |
              (!is.na(vbg_co2) & !is.na(vbg_ph))
  ) %>%
  filter(
    # Keep strict HRF cases
    hrf_hypercapnic_v2_strict == 1 |

    # OR clean controls with labs and no HRF by any definition
    (
      has_gas == TRUE &
      hrf_hypercapnic_labs_v2 == 0 &
      hrf_hypercapnic_icd_v2  == 0
    )
  )
```

```
model_strict_bmi_unadj <- glm(
  hrf_hypercapnic_v2_strict ~ curr_bmi_num,
  family = binomial,
  data = stata_hrf_strict_analysis
)

summary(model_strict_bmi_unadj)
```

```
Call:
glm(formula = hrf_hypercapnic_v2_strict ~ curr_bmi_num, family = binomial,
    data = stata_hrf_strict_analysis)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -3.555387   0.060186  -59.07   <2e-16 ***
curr_bmi_num  0.023892   0.001946   12.28   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 39142  on 92391  degrees of freedom
Residual deviance: 38995  on 92390  degrees of freedom
  (149227 observations deleted due to missingness)
AIC: 38999

Number of Fisher Scoring iterations: 5
```

```r
or_table <- exp(cbind(
  OR = coef(model_strict_bmi_unadj),
  confint(model_strict_bmi_unadj)
))
```

```
Waiting for profiling to be done...
```

```r
or_table
```

```
                    OR      2.5 %     97.5 %
(Intercept)  0.02857031 0.02538715 0.03214267
curr_bmi_num 1.02417964 1.02027437 1.02808618
```

Then adjusted for age, sex, ethnicity:

```r
baseline_model_strict <- glm(
  hrf_hypercapnic_v2_strict ~
    curr_bmi_num +
    age_at_encounter +
    sex +
    ethnicity,
  data = stata_hrf_strict_analysis,
  family = binomial
)

summary(baseline_model_strict)
```

```
Call:
glm(formula = hrf_hypercapnic_v2_strict ~ curr_bmi_num + age_at_encounter +
    sex + ethnicity, family = binomial, data = stata_hrf_strict_analysis)

Coefficients:
                             Estimate Std. Error z value Pr(>|z|)
(Intercept)                -4.9237097  0.0957773 -51.408  < 2e-16 ***
curr_bmi_num                0.0279252  0.0019885  14.043  < 2e-16 ***
age_at_encounter            0.0203174  0.0009537  21.304  < 2e-16 ***
sexMale                    -0.0419128  0.0292192  -1.434  0.15145
ethnicityHispanic or Latino -0.1985303  0.0634460  -3.129  0.00175 **
ethnicityUnknown            0.0143449  0.0424710   0.338  0.73555
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 39142  on 92391  degrees of freedom
Residual deviance: 38462  on 92386  degrees of freedom
  (149227 observations deleted due to missingness)
AIC: 38474

Number of Fisher Scoring iterations: 6
```

```r
exp(cbind(
  OR = coef(baseline_model_strict),
  confint(baseline_model_strict)
))
```

```
Waiting for profiling to be done...
```

	OR	2.5 %	97.5 %
(Intercept)	0.007272104	0.006022747	0.008766983
curr_bmi_num	1.028318745	1.024313036	1.032328880
age_at_encounter	1.020525190	1.018627045	1.022442335
sexMale	0.958953425	0.905590654	1.015494380
ethnicityHispanic or Latino	0.819934924	0.722589291	0.926707324
ethnicityUnknown	1.014448311	0.932744796	1.101733222

Chi square analysis, just descriptive:

```r
# Subset to patients with BMI categories
bmi_hrf_strict_subset <- stata_hrf_strict_analysis %>%
  filter(!is.na(bmi_category))

# Summary table by BMI category
bmi_hrf_strict_summary <- bmi_hrf_strict_subset %>%
  group_by(bmi_category) %>%
  summarise(
    n_total = n(),
    n_hrf = sum(hrf_hypercapnic_v2_strict == 1),
    n_no_hrf = sum(hrf_hypercapnic_v2_strict == 0),
    pct_hrf = round(100 * n_hrf / n_total, 1),
    pct_no_hrf = round(100 * n_no_hrf / n_total, 1)
  ) %>%
  arrange(factor(bmi_category, levels = c("Underweight", "Normal weight", "Overweight",
                                          "Obesity I", "Obesity II", "Obesity III")))
```

```r
# Chi-square test
chisq_test_strict <- chisq.test(table(
  bmi_hrf_strict_subset$bmi_category,
  bmi_hrf_strict_subset$hrf_hypercapnic_v2_strict
))

# View summary table and p-value
bmi_hrf_strict_summary
```

bmi_category	n_total	n_hrf	n_no_hrf	pct_hrf	pct_no_hrf
Underweight	4054	369	3685	9.1	90.9
Normal weight	25881	1246	24635	4.8	95.2
Overweight	26633	1110	25523	4.2	95.8
Obesity I	17896	870	17026	4.9	95.1
Obesity II	9917	638	9279	6.4	93.6
Obesity III	8011	811	7200	10.1	89.9

```r
chisq_test_strict$p.value
```

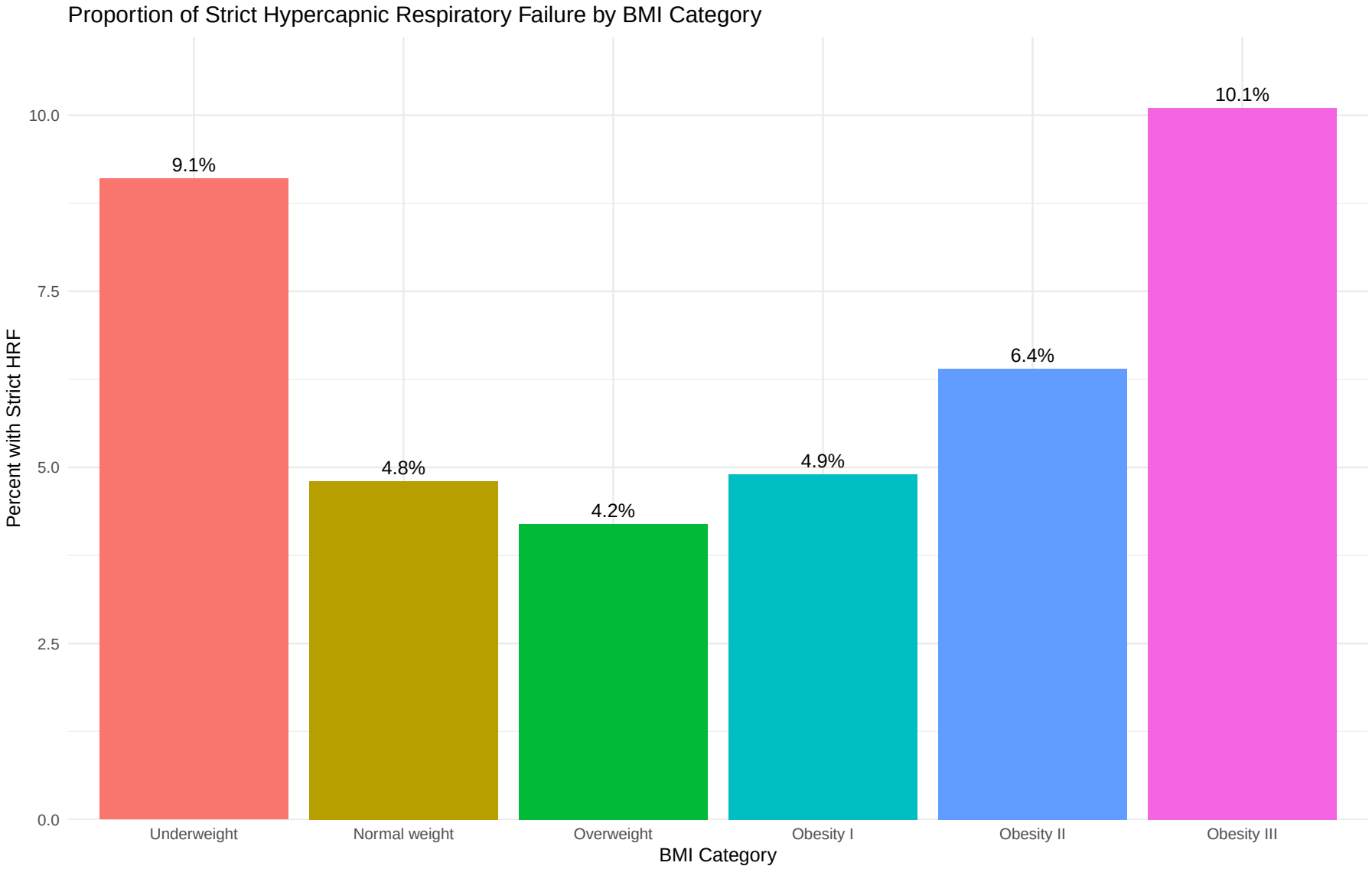
```
[1] 5.52785e-123
```

```r
library(ggplot2)

# Make sure bmi_category is ordered
bmi_hrf_strict_summary <- bmi_hrf_strict_summary %>%
  mutate(bmi_category = factor(bmi_category,
                               levels = c("Underweight", "Normal weight", "Overweight",
                                          "Obesity I", "Obesity II", "Obesity III")))

# Plot proportion of strict HRF by BMI category
ggplot(bmi_hrf_strict_summary, aes(x = bmi_category, y = pct_hrf, fill = bmi_category)) +
  geom_col() +
  geom_text(aes(label = paste0(pct_hrf, "%")), vjust = -0.5) +
```

```
  scale_y_continuous(expand = expansion(mult = c(0, 0.1))) +
  labs(
    title = "Proportion of Strict Hypercapnic Respiratory Failure by BMI Category",
    x = "BMI Category",
    y = "Percent with Strict HRF"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```


Proportion of Strict Hypercapnic Respiratory Failure by BMI Category
Percent with Strict HRF
10.0
7.5
5.0
2.5
0.0
9.1%
4.8%
4.2%
4.9%
6.4%
10.1%
Underweight
Normal weight
Overweight
Obesity I
Obesity II
Obesity III
BMI Category